

Staphylococcus aureus Bacteremia — The 5 Core Interventions

SAB is never a contaminant · ID consult + echo + repeat cultures + source control + right drug/duration · MSSA: cefazolin/antistaph penicillin (NOT vancomycin) · never treat as a simple line UTI

1 · Take it seriously — the 5 core interventions

A positive *S. aureus* blood culture is **never** a contaminant. Five evidence-based interventions reduce mortality — do all five, every time:

#	Core intervention
1	ID consultation — lowers mortality
2	Echocardiography to evaluate for endocarditis
3	Repeat blood cultures q48–72h to document clearance
4	Source control — remove lines, drain foci
5	Appropriate drug & duration — set by complicated vs uncomplicated

ID consultation independently improves survival in SAB — involve them in every case, not just the complicated ones.

2 · Right drug — MSSA vs MRSA

- **Start empirically with vancomycin** while awaiting susceptibilities (covers MRSA).
- **MSSA: de-escalate to cefazolin or an antistaphylococcal penicillin** (nafcillin/oxacillin) — both beat vancomycin for MSSA. **Vancomycin is inferior for MSSA** — do not 'leave it on' once you know it's methicillin-susceptible.
- Cefazolin is convenient and well-tolerated; antistaph penicillins avoid the theoretical inoculum effect in deep infection.
- **MRSA:** vancomycin (target AUC-guided) or daptomycin.

The classic error: a patient with MSSA bacteremia left on vancomycin for the whole course. Susceptible → switch to cefazolin/nafcillin.

3 · Complicated vs uncomplicated → duration

Duration hinges on this split. **Uncomplicated** SAB requires **all** of:

Uncomplicated needs ALL	Complicated if ANY
Negative follow-up cultures (clearance)	Persistent positive cultures
Defervescence within 72 h	Endocarditis
No endocarditis / no prosthetic material	Implanted prosthesis/hardware
No metastatic foci	Metastatic infection (bone, joint, abscess)

Uncomplicated SAB: **≥14 days** IV from the first negative culture.
Complicated SAB: **4–6 weeks** (longer for endocarditis/osteomyelitis). Count the clock from clearance, not from day one.

Hunt for metastatic foci

- *S. aureus* seeds widely: **endocarditis, vertebral osteomyelitis/discitis, epidural abscess, septic arthritis, psoas/splenic abscess, septic emboli**.
- Ask about and exam for back pain (osteo/epidural — image with MRI), joint pain, new murmur, peripheral stigmata.
- **TEE** is more sensitive than TTE for endocarditis; pursue TEE in higher-risk or persistent cases (prosthetic valve/device, persistent bacteremia, community-onset, no clear source).

Do/Avoid

- **DO** consult ID, get an echo, repeat cultures, control the source, and set drug/duration on every SAB
- **DO** de-escalate MSSA to cefazolin or an antistaph penicillin
- **DO** time the duration from the first negative culture
- **AVOID** calling *S. aureus* a contaminant or treating it like a simple line infection
- **AVOID** keeping a confirmed MSSA on vancomycin
- **AVOID** a short course without confirming uncomplicated criteria (clearance, no endocarditis, no foci)